

Grafana ↔ Jasper Parity — Response to Testing Feedback (2026-06-25)

Covers the three reports flagged in *Grafana Reports Testing.pdf* for the 6/18 12am → 6/19 12am comparison window.

TL;DR

Two of the three "mismatches" were **not dashboard bugs** — they were caused by the Grafana time picker being on a **different 24-hour window** than the Jasper report (a timezone effect, explained below). When both are pointed at the *same* window, the numbers match Jasper exactly. The third (Alarm Log) had a small, genuine query-logic difference that has now been fixed.

Report	Status	Detail
Alarm Activity by User	✅ Already correct	At Jasper's window = 400 / 0 / 0 / 0 / 0 — exact match. The 427 was the shifted window.
Top-N Most Common Alarms	✅ Already correct	At Jasper's window: total 413 , top type 78 — exact match.
Alarm Log	✅ Fixed (v7)	Now matches Jasper's "active-during-window" counting. Total 360 .

The main cause: time zone of the dashboard time picker

- The reporting database stores all alarm times in **UTC**.
- The Jasper reports use **UTC midnight → UTC midnight** as the day boundary

(verified: querying that exact window reproduces every Jasper number).

- The Grafana dashboards are set to **Browser time**, meaning the time picker shows and accepts times in **your computer's local time zone**.

So if your browser is, say, US Eastern (UTC-4) and you pick "midnight," Grafana

actually queries a window shifted by 4 hours from the one Jasper used — which pulls in (or drops) alarms near the boundary and makes the totals differ. This is exactly what produced the "extra" 27 activities in Alarm Activity and the Top-N differences. The dashboard math was right; the **window was different**.

How to line Grafana up with Jasper (end-user steps)

You can switch any dashboard's view to UTC yourself — no admin change needed:

1. Click the **time-range** control (top-right of the dashboard).
2. Click **Change time settings**.
3. Set **Time zone** → **Coordinated Universal Time (UTC)**.
4. Now enter the same window Jasper uses, e.g.

2026-06-18 00:00:00 to 2026-06-19 00:00:00 .

With the picker in UTC, the displayed times and the queried window match

Jasper's report period one-for-one, and the counts line up.

Note: in Browser-time mode the timestamps shown in tables are in *your* local

zone, so an alarm Jasper lists at 08:09 PM may display at a different

clock time for you even though it is the same event. Switching the picker to

UTC also makes the displayed timestamps match Jasper.

Alarm Log — the one genuine fix (dashboard v7)

Jasper's Alarm Log counts every alarm **active at any point during the window**

(opened before and cleared inside it, opened inside and still open, etc.), and

applies its minimum-duration filter **only to alarms that have cleared**.

Our dashboard previously counted only alarms **opened inside** the window and

applied the duration filter to every alarm. That under-counted long-running

alarms that spanned the boundary.

Fixed: the count tiles and the Alarm Log table now use Jasper's active-during-window logic and cleared-only minimum-duration. Over the 6/18 (UTC) window this yields:

Tile	Value
Total Alarms	360
Critical	63
Major	297
Minor	0
Devices w/ Alarms	114

(The "Alarms per Day" chart intentionally still buckets by the alarm's *start day*, so its per-day bars can differ slightly from the active-during-window total — that is expected for a daily trend chart.)

What was checked, for the record

- `alarm_key` is unique in `alarminfo` (no double-counting from joins).
- Raw `alarmactivity` for UTC 6/18 = 400 cleared, 0 others → matches Jasper.
- Top-N: condition filter makes no difference for this data (all alarms are

Critical/Major/Minor); total 413 and top type 78 match Jasper.

- All numbers reproduced directly against the reporting database and confirmed on the live dashboard.